

LOMBA KOMPETENSI SISWA (LKS) SEKOLAH MENENGAH KEJURUAN TINGKAT PROVINSI JAWA TENGAH TAHUN 2023

KISI-KISI SOAL



Bidang Lomba IT NETWORK SYSTEMS ADMINISTRATION



PEMERINTAH PROVINSI JAWA TENGAH DINAS PENDIDIKAN

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INFORMASI UMUM

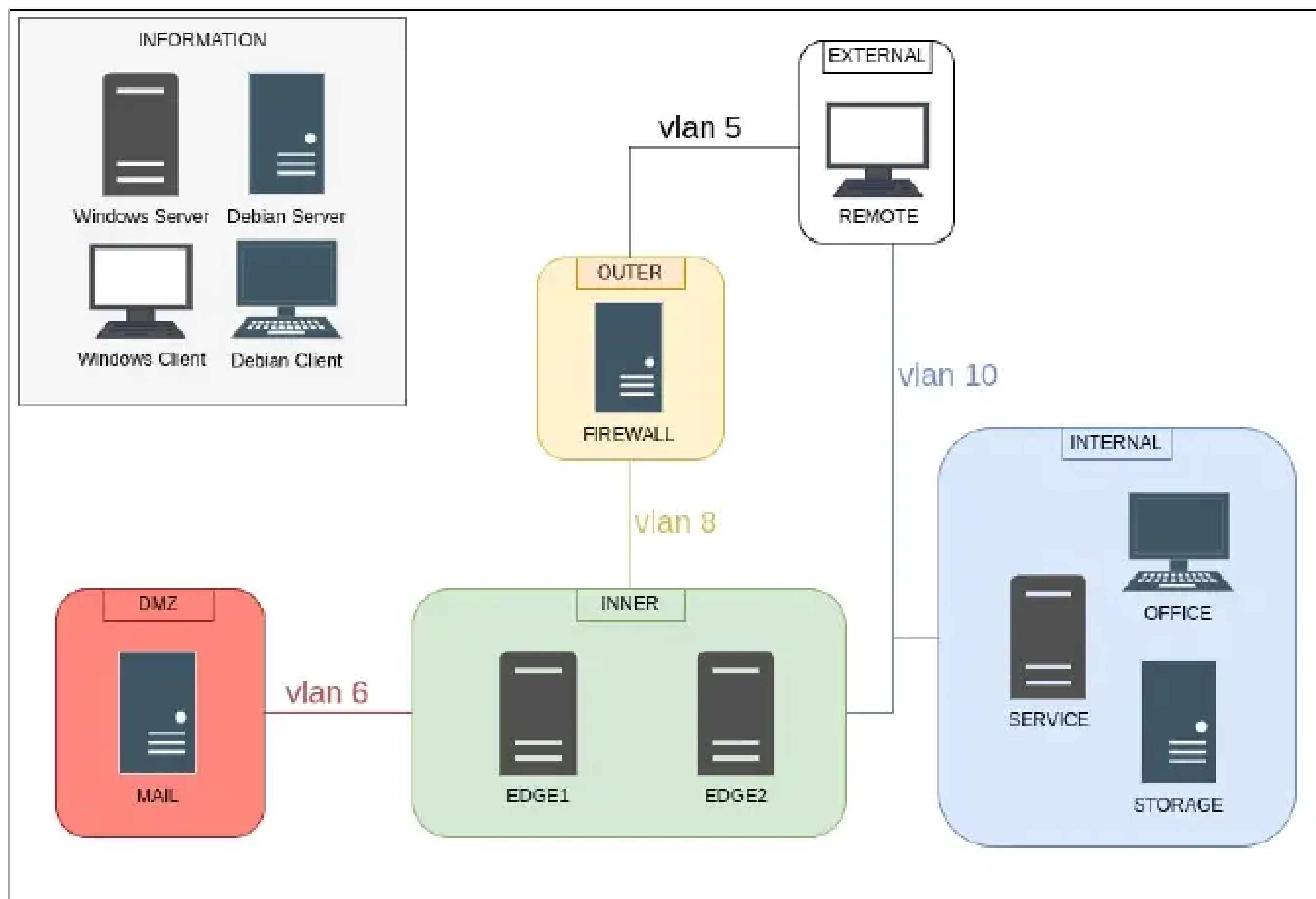
Lomba Kompetensi Siswa (LKS) tingkat Provinsi Jawa Tengah tahun 2023 bidang lomba IT Network Systems Administration (ITNSA) menggunakan Test Project yang mengacu kepada Test Project LKS Tingkat Nasional 2022, maka kisi-kisi yang diberikan merupakan kisi-kisi berdasarkan Test Project LKS Nasional tahun 2022. Berkaitan dengan hasil tersebut, maka Test Project Actual yang akan digunakan pada pelaksanaan LKS Tingkat Provinsi tahun 2023 akan dibuka pada saat hari kompetisi dengan memberikan perubahan maksimal 30% setiap modul. Perubahan tersebut meliputi topology, IP Address, nama host dan domain serta termasuk perubahan perintah spesifik yang diberikan.

MODULE A

CLIENT SERVER ENVIRONMENT

A branch of a big company has hired you to configure their newly built infrastructure. There will be three sites, DMZ, Internal, and Edge. They will use both Windows and Linux for their services. Attached the logical topology of the new branch infrastructure.

Topology **(Update: 15 Mei 2023)**



Client Server

- **Edge**
 - Firewall
 - Load Balancing Traffic
 - Routing
 - Routing Failover
 - Configure DHCP
 - Network Boot
- **DMZ**
 - Configure Service Monitoring
 - Configure Mail Service
 - Configure LDAP
 - Service Failover
 - Remote Access VPN
 - Certificate Authority

- **Internal**
 - Shared File Service
 - Configure Web Service
 - Enable HTTPS
 - Configure DNS
 - Backup and Restore
 - Role Based Access Policy
 - Configure Operating System

Configuration Table

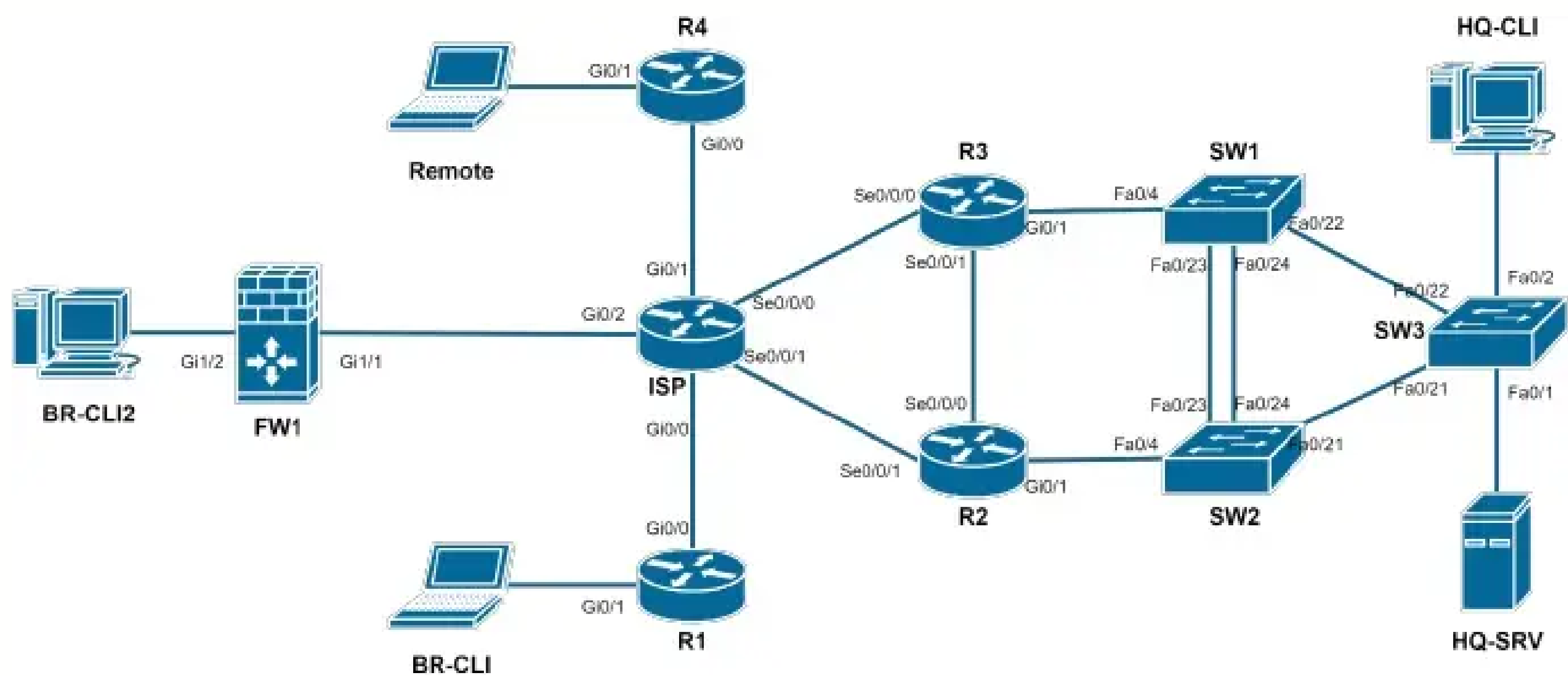
Hostname	Operating System	IP Address	Preinstalled
EDGE1	Windows Server 2019 desktop	172.16.234.253/24	yes
		172.16.233.125/25	
		172.19.99.252/28	
EDGE2	Windows Server 2019 desktop	172.16.234.254/24	yes
		172.16.233.126/25	
		172.19.99.253/28	
SERVICE	Windows Server 2019 desktop	172.16.233.11/25	yes
FIREWALL	Debian 11 Server	172.19.99.254/28	yes
		172.12.33.25/30	
REMOTE	Windows 10	172.16.233.10/25	yes
		172.12.33.26/30	
MAIL	Debian 11 Server	172.16.234.21/24	yes
STORAGE	Debian 11 Server	172.16.233.15/25	yes
OFFICE	None	DHCP	no

MODULE B

NETWORK TROUBLESHOOTING – CISCO PACKET TRACER

Kemendikbud has a complicated network infrastructure to connect Workstations and Servers in different places. Network traffic will be managed using the technologies of Cisco devices including Layer-3 Switches, Regular Switches, and Routers. Please configure according to the requirements to enable reliable connection between different places.

Logical Topology

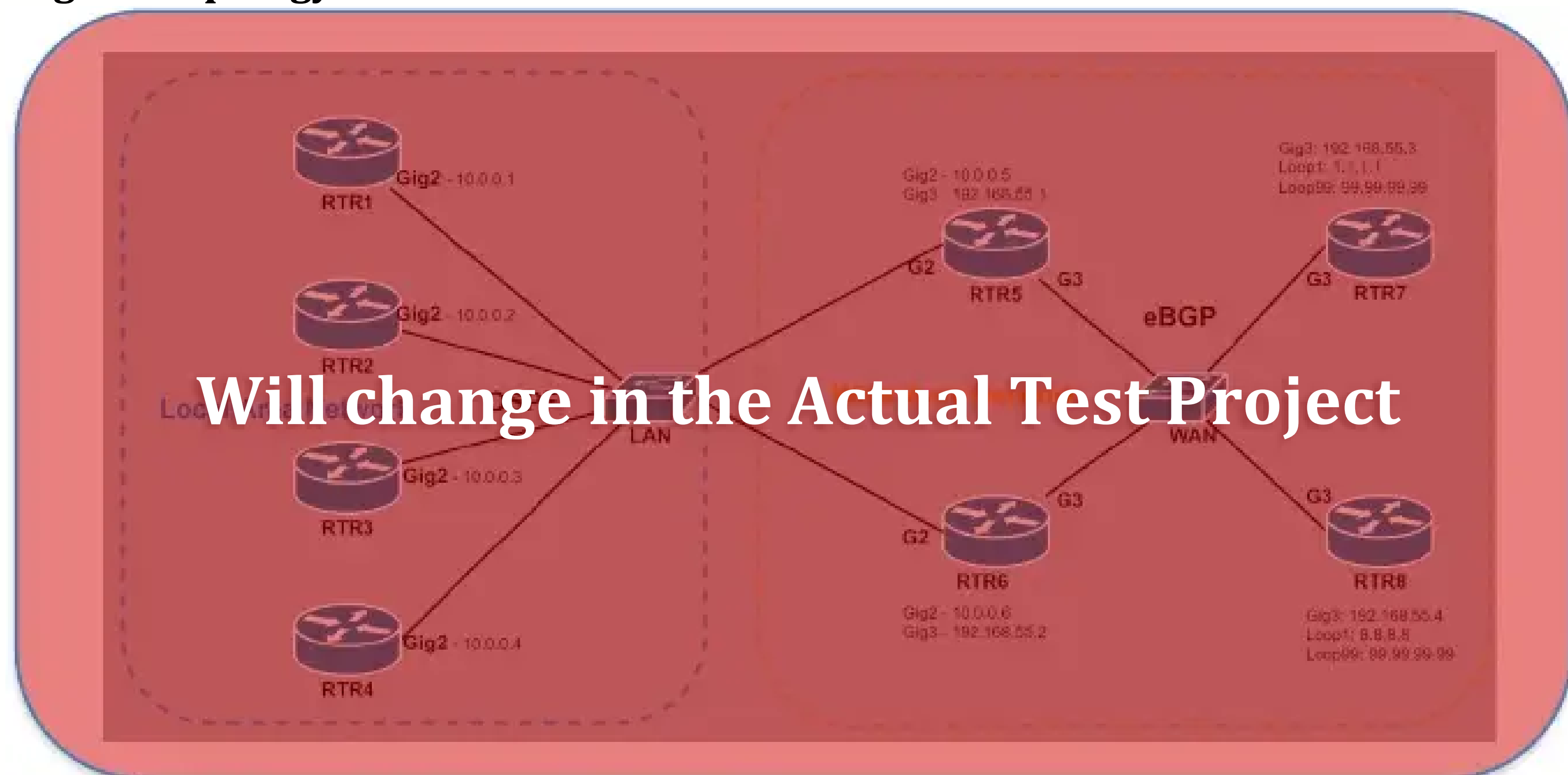


MODULE C

NETWORK SYSTEMS – CISCO MODELING LAB (CML)

Modul C Network Systems ini akan menggunakan Cisco Modeling Lab (CML). Peserta akan diminta untuk mengkonfigurasi Network Routing & Switching. Terdapat dua area, Local Area Network & Wide Area Network.

Logical Topology



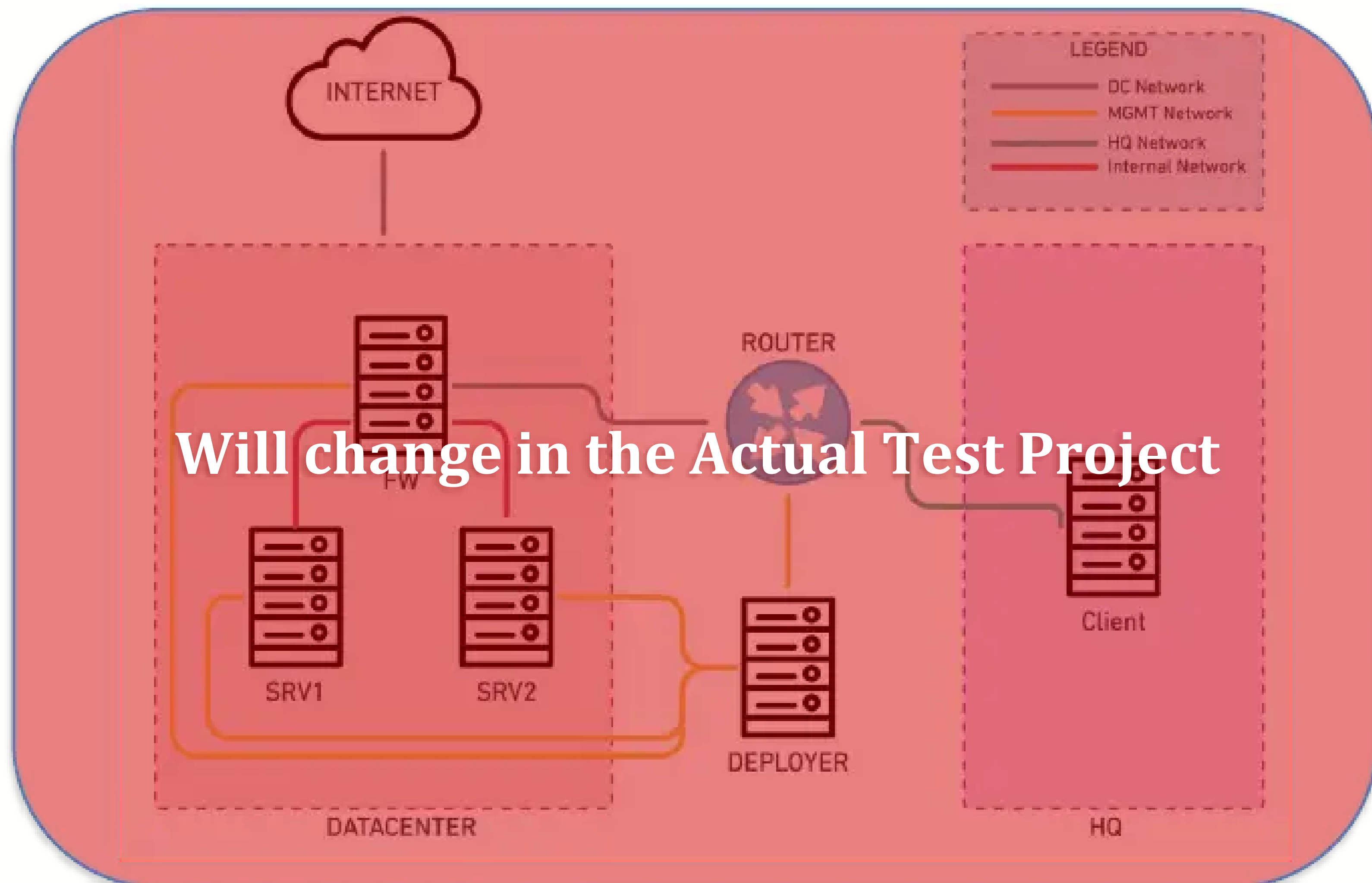
- **Task Summery**
 - Basic Configuration & IP Addressing
 - Static Routing
 - Dynamic Routing Protocol
 - LAN Switching
 - Virtual Private Network
 - Network Security
 - Network Address Translation

MODULE D

INFRASTRUCTURE PROGRAMMABILITY & AUTOMATION

A small UMKM in Indonesia just procured a lot of servers to support their containerized online services. They are hiring for new engineers to help manage the servers. This is the sample of their technical test for recruitment. You will need to utilize many tools to configure, manage, and deploy the infrastructure.

Topology



INFRASTRUCTURE PROGRAMMABILITY & AUTOMATION

- **Network Programmability**
 - RESTCONF
 - Python
- **Ansible**
 - Create Playbook
 - Network Services
 - System Admin (ex. webserver)
- ~~**Containerized Services**~~
 - ~~Dockerize Python Application~~
 - ~~Running Containers~~

Configuration Table

Device	IP Address	OS
router	172.16.10.1/24 (DC) 192.168.10.1/24 (HQ) 10.10.10.10/24 (MGMT)	CSR 1000v 16.9.5
deployer	10.10.10.1/24 (MGMT) DHCP (inet)	Debian 11.x
FW	172.16.10.2/24 (DC) DHCP (inet) 10.10.10.2/24 (MGMT) 172.16.20.1/24 (Internal)	Debian 11.x
srv1	172.16.20.10/24 (Internal) DHCP (inet) 10.10.10.3/24 (MGMT)	Debian 11.x
srv2	172.16.20.20/24 (Internal) DHCP (inet) 10.10.10.4/24 (MGMT)	Debian 11.x
client	192.168.10.100/24 (HQ)	Debian 11.x